

Lab Assignment 9

MTH 308: Numerical Analysis and Scientific Computing-I

Department of Mathematics and Statistics, IIT Kanpur

2026 Even Semester: April 17, 2026

In this lab, you will implement composite numerical quadrature rules. Please write a single script file named `Lab9LastnameRollNumber.m` and publish it as a PDF. Then submit the script `Lab9LastnameRollNumber.m` along with the corresponding published file `Lab9LastnameRollNumber.pdf` (for example, my submitted files would be `Lab9Biswas260102.m` and `Lab9Biswas260102.pdf`) to HELLO IITK.

Each solution should:

- Be contained in a separate cell, including the problem number and a short problem description.
- Be adequately commented to explain your logic.
- Run independently of other cells.

1. Approximate

$$\int_e^{e+2} \frac{1}{x \ln x} dx$$

- (a) using the composite trapezoidal rule with $n = 6$,
- (b) using the composite Simpson's rule with $n = 6$.

Compare your results with the exact value $\ln(\ln(e+2))$.

Composite Trapezoidal Rule. Let $f \in C^2[a, b]$, $h = \frac{b-a}{n}$, and $x_i = a + ih$ for $i = 0, 1, \dots, n$. Then there exists $\mu \in (a, b)$ such that the composite trapezoidal rule can be written as

$$\int_a^b f(x) dx = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right] - \frac{b-a}{12} h^2 f''(\mu).$$

Composite Simpson's Rule. Let $f \in C^4[a, b]$, n be even, $h = \frac{b-a}{n}$, and $x_i = a + ih$ for $i = 0, 1, \dots, n$. Then there exists $\mu \in (a, b)$ such that the composite Simpson's rule can be written as

$$\int_a^b f(x) dx = \frac{h}{3} \left[f(a) + 2 \sum_{i=1}^{\frac{n}{2}-1} f(x_{2i}) + 4 \sum_{i=1}^{\frac{n}{2}} f(x_{2i-1}) + f(b) \right] - \frac{b-a}{180} h^4 f^{(4)}(\mu).$$